

## **Guidelines for Tutorial #2**

### **Unsupervised Machine Learning**

#### **Building a Recommendation System**

## **1 Introduction**

This tutorial challenges you to apply advanced unsupervised learning techniques to build and refine a recommendation system. You will begin by selecting a suitable dataset, using the sources available in Chapter 4 of this document. Data wrangling and visualization are essential initial steps to ensure data quality. Your main task is to integrate techniques like outlier detection, rule association mining, and self-organizing maps into your recommendation system. Additionally, you are required to compare these techniques with clustering methods to explore different aspects of your dataset and refine your recommendation logic.

## **2 Recommendation Systems**

A recommendation system (or recommender system) is a class of machine learning that uses data to help predict, narrow down, and find what people are looking for among an exponentially growing number of options. These can be based on various criteria, including past purchases, search history, demographic information, and other factors. Recommender systems are highly useful as they help users discover products and services they might otherwise have not found on their own.

Recommender systems are trained to understand the preferences, previous decisions, and characteristics of people and products using data gathered about their interactions. These include impressions, clicks, likes, and purchases. Because of their capability to predict consumer interests and desires on a highly personalized level, recommender systems are a favorite with content and product providers. They can drive consumers to just about any product or service that interests them, from books to videos to health classes to clothing.

## 2.1 Use Cases

- **e-Commerce:** AI-based recommendation systems are widely used in e-commerce to suggest products to users based on their browsing and purchase history, preferences, and behavior. The recommendation system analyzes customer data to identify patterns and trends in their purchase behavior and recommends products that interest them.
- **Supply Chain:** AI-based recommendation systems can be used in supply chain management to optimize inventory and reduce waste. The recommendation system can analyze historical inventory levels, sales data, and supplier lead times to forecast demand and suggest optimal ordering and inventory management strategies.
- **Media and Entertainment:** Based on their viewing and listening history, preferences, and behavior, recommendation systems suggest movies, TV shows, and music to users. The recommendation system analyzes the user's viewing and listening history to identify patterns and preferences and recommends content likely to interest them.
- **Healthcare:** Recommendation systems are used in healthcare to suggest treatments and medications to patients based on their medical history, symptoms, and other relevant data. The recommendation system analyzes patient data to identify patterns and trends in their medical history and recommends treatments that are likely to be effective.
- **Financial Services:** Recommendation systems are used in financial services to suggest financial products to customers based on their financial history, behavior, and preferences. The recommendation system analyzes customer data to identify patterns and trends in their financial history and recommends products likely to meet their financial needs.
- **Travel and Accommodation:** Recommendation systems are used in the travel and hospitality industry to suggest hotels, flights, and other travel-related services to users based on their preferences, behavior, and budget. The recommendation system analyzes customer data to identify patterns and trends in their travel history and recommends services likely to meet their needs.

## 3 Requirements

- Pick a dataset suitable for building a recommendation system by selecting from the sources below. Consider datasets that allow for the application of unsupervised learning techniques and have potential for insightful data associations.
- If the selected dataset contains issues, it must be cleaned or "wrangled" using the Pandas library in Python.
- **Develop and implement** a recommendation system using:

- A rule association mining with **Apriori algorithm** (and / or others) AND / OR:
- A **Self-Organizing Map** (SOM) for clustering / dimensionality reduction.
- **Compare** the effectiveness of your recommendation system when using:
  - Clustering techniques previously learned (k-Means, Hierarchical Clustering, DBSCAN) AND / OR
  - SOMs vs. other clustering methods.

This comparison should focus on the quality of recommendations, and any insights gained that could improve or refine the recommendation logic.

- Document the implementation, comparison, and insights gained from applying these unsupervised learning techniques in a Jupyter Notebook.
- Ensure the Jupyter Notebook is well-organized, with clear comments and adheres to best coding practices.

## 4 Data Acquisition

There are some repositories where you can search for public datasets and use them for your recommendation system:

| # | Platform / Repository            | URL                                                                                                                         |
|---|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | Data.world                       | <a href="https://data.world/">https://data.world/</a>                                                                       |
| 2 | Kaggle Datasets                  | <a href="https://www.kaggle.com/datasets">https://www.kaggle.com/datasets</a>                                               |
| 3 | Awesome Public Datasets - GitHub | <a href="https://github.com/awesomedata/awesome-public-datasets">https://github.com/awesomedata/awesome-public-datasets</a> |
| 4 | UCI Machine Learning Repository  | <a href="https://archive.ics.uci.edu/ml">https://archive.ics.uci.edu/ml</a>                                                 |
| 5 | OpenML                           | <a href="https://openml.org/">https://openml.org/</a>                                                                       |
| 6 | Amazon Datasets                  | <a href="https://registry.opendata.aws/">https://registry.opendata.aws/</a>                                                 |
| 7 | Google's Datasets Search Engine  | <a href="https://toolbox.google.com/datasetsearch">https://toolbox.google.com/datasetsearch</a>                             |

## 5 Group Composition

- Tutorials will be conducted in groups of 3-4 elements.
- However, evaluations will be carried out on an **individual** basis.

## 6 Project Submission

- Submit the final Jupyter Notebook (.ipynb) with your implementations and findings, plus the dataset that you used to accomplish this tutorial.
- The deadline for final submission on Moodle is **June 1st, 23h59 UTC**.

## 7 Oral Discussion

- The oral discussion will take place on **June 2nd**, from **09h30** to **12h30**.
- Your group needs to choose your time slot in this [link](#).
- Only one group member, the group representative, should select the day and time slot suitable for the entire group from the options provided. This selection will also be on a first-come, first-served basis.
- To minimize downtime between sessions, opt for the **earliest** available time slot.
- Each discussion will last for **20 minutes**, during which the lecturer may ask **questions** about your project.
- Please arrive **5-10 minutes** prior to your scheduled time to prevent delays.
- Only one member from each group needs to **bring a laptop** with the notebook **prepared and ready for demonstration**.
- Grades **will not be provided** immediately after the discussion. Instead, all projects will be reviewed comprehensively before grading.

## 8 Evaluation Criteria

- **CA1:** Correctness of Data Wrangling and preparation using Pandas (if applicable).
- **CA2:** Comprehensive application and interpretation of the Apriori algorithm for rule association mining, including a detailed analysis of the rules discovered and their relevance to the recommendation logic **AND / OR** the use of Self-Organizing Maps for clustering.
- **CA3:** Correctness of the comparison between previously learned techniques and the new ones.
- **CA4:** In-depth comparative analysis of the developed recommendation system's performance.
- **CA5:** Clarity, organization, and documentation of the analytical process.
- **CA6:** Responsiveness and clarity in addressing any queries from the lecturer.

## 9 Additional Information

- Maximum grade: 20 points.
- Weight: 30%.